

Multiple Choice (10 marks)

1. When the Heisenberg uncertainty principle is applied to a quantum mechanical particle in the lowest energy level of a one-dimensional box, which of the following is true?
 - (a) Momentum is known exactly, but no information about position can be known.
 - (b) Position is known exactly, but no information about momentum can be known.
 - (c) No information about either position or momentum can be known.
 - (d) Neither position nor momentum can be known exactly.
2. Which of the following statements most accurately explains why the ^1H spectrum of $^{12}\text{CHCl}_3$ is a singlet?
 - (a) Both ^{35}Cl and ^{37}Cl have $I = 0$.
 - (b) The hydrogen atom undergoes rapid intermolecular exchange.
 - (c) The molecule is not rigid.
 - (d) Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.
3. Which one of the following statements is true:
 - (a) Protons and neutrons have orbital and spin angular momentum.
 - (b) Protons have orbital and spin angular momentum, neutrons have spin angular momentum.
 - (c) Protons and neutrons possess orbital angular momentum only.
 - (d) Protons and neutrons possess spin angular momentum only.
4. Which of the following is classified as a conjugate acid-base pair?
 - (a) HCl / NaOH
 - (b) $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$
 - (c) $\text{O}_2 / \text{H}_2\text{O}$
 - (d) H^+ / Cl^-
5. Which of the following statements about the lanthanide elements is NOT true?
 - (a) The most common oxidation state for the lanthanide elements is +3.
 - (b) Lanthanide complexes often have high coordination numbers (> 6).
 - (c) All of the lanthanide elements react with aqueous acid to liberate hydrogen.
 - (d) The atomic radii of the lanthanide elements increase across the period from La to Lu.

True / False (10 marks)

Answer True or False

6. True or False: CH_3F shows a ^1H NMR resonance at a similar frequency as CH_3Cl when measured in a 12.0 T magnetic field.
7. True or False: The magnetic field responsible for the polarization of 2.5×10^{-6} for ^{13}C at 298 K is 100 T.
7. True or False: Unpaired electrons are essential for both paramagnetism and ferromagnetism in materials.
8. True or False: Lead is often deemed a safe substance in aquatic environments.
9. True or False: Ammonia, NH_3 , is the strongest base in liquid ammonia.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the orbital angular momentum quantum number, l , in determining the electron that is most easily removed from ground-state aluminum during ionization.
12. Discuss the factors that contribute to the difference in first ionization energy between argon and neon, and explain why argon's first ionization energy is lower.

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